

# Gannavarapu Gokul Madhav

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## SUMMARY

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Aspiring AI/ML Engineer and Full-Stack Developer with hands-on experience in deep learning, NLP, and computer vision. Published researcher (JTAIT 2025) and Research Intern at NIT Rourkela. Proficient in Python, PyTorch, Java, and web technologies.

## SKILLS

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AI / ML & NLP, YOLOv5, EfficientNet, ResNet, Longformer, PyTorch, CNN, TensorRT, Transfer Learning, Python, Java, JavaScript, PHP, HTML5, CSS, Flask, MySQL, Git, OpenCV, Add 'Agentic AI' to the Skills section., Add 'Generative AI' to the Skills section., Add 'LLM' to the Skills section., Add 'Machine Learning' to the Skills section., Add 'NLP' to the Skills section., Add 'Artificial Intelligence' to the Skills section.

## EXPERIENCE

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### Research Intern – 3D Face Reconstruction — NIT Rourkela (July 2025 – August 2025)

Built a 3D face reconstruction pipeline using DECA and Poisson Surface Reconstruction techniques. Implemented RANSAC-based landmark refinement to generate high-fidelity 3D meshes from single 2D images.

- Expand the Experience section to include specific projects or roles that utilized AI/ML, NLP, and other relevant skills.

## EDUCATION

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**B.Tech** in Computer Science & Engineering (AI & ML) — Lakireddy Bali Reddy College of Engineering (2026)

**Diploma** in Computer Science & Engineering — Sree Vahini Institute of Science and Technology (2023)

## PROJECTS

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### Legal Document Analyzer

Built a long-context legal text classifier using domain-adapted Longformer on 26,000+ Indian Supreme Court judgments (1950–2025) to predict outcomes (Allowed, Dismissed, Transferred, Remanded). Achieved 90% accuracy and 0.85 macro F1-score; applied class-weighted training and token-length experiments (1024–4096) showing improved performance with longer context.

### Mosquito Classification

Achieved ~90% accuracy on a custom 1,000+ image dataset; published in JTAIT Vol. 103 (2025) with 92.3% accuracy. Leveraged CSPDarknet53 for feature extraction and PANet for multi-scale analysis; optimized hyperparameters for precision.

### Real-Time Brain Tumor Classification from MRI

Lightweight 3D CNN on BraTS 2021; achieved 87% Dice score and 18 FPS on RTX 4060 via pruning and quantization. Applied model compression (pruning, quantization, TensorRT) to enable real-time inference on edge hardware.

### Skin Disease Classification – HAM10000

Ensemble of EfficientNet-B5, ResNet-101, and custom CNN classifying 7 lesion types into 3 categories; 89% test accuracy. Applied transfer learning and data augmentation on HAM10000 to handle class imbalance effectively.

### 3D Face Reconstruction from 2D Images

Developed an end-to-end pipeline using DECA (Detailed Expression Capture and Animation) from single 2D images. Integrated RANSAC-based 3D landmark refinement and Poisson surface reconstruction for high-quality mesh generation.

## **Crackerwala – E-Commerce Platform**

Built a full-stack shopping site (crackerwala.in) with product listings, cart, and payment integration. Designed MySQL schema for inventory and order management; reduced checkout time by 30% via optimized SQL queries.